

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

Consider the central planner problem from the previous homework:

$$\begin{aligned} \max E \sum_{t=0}^{\infty} \beta^t & \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\xi L_t^{1+\eta^{-1}}}{1+\eta^{-1}} \right\} \\ \text{subject to} & \\ Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha} & \\ Y_t = C_t + I_t + \frac{\phi}{2} \left( \frac{I_t}{K_{t-1}} - \delta \right)^2 K_{t-1} & \\ K_t = (1-\delta)K_{t-1} + I_t & \\ Z_t = e^{\varepsilon_t} Z_{t-1}^\rho & \end{aligned}$$

Assume that  $\rho < 1$  and  $\varepsilon_t \sim iid(0, \sigma_\varepsilon^2)$ . The structural parameters of the model are  $(\sigma, \alpha, \xi, \eta, \delta, \rho, \phi, \beta, \sigma_\varepsilon)$ . Suppose that  $\alpha = 1/3, \eta = 1, \phi = 5, \delta = 0.025, \beta = 0.99, \sigma = 1, \rho = 0.9, \sigma_\varepsilon = 1$ , and  $\xi = 1$ . Suppose that you know  $(\sigma, \alpha, \xi, \delta, \beta, \eta)$  and you want to estimate  $\theta = (\rho, \phi, \sigma_\varepsilon)$ .

Suppose that observed variables  $x_t = [c_t \ i_t \ y_t \ l_t]'$  are measured with i.i.d. noise so that the series available to the econometrician are  $x_t^* = x_t + e_t$  with  $e_t \sim iidN(0, \Sigma)$  with the covariance matrix of measurement errors being diagonal  $\Sigma = diag\{\sigma_e^2, \sigma_e^2, \sigma_e^2, \sigma_e^2\}$ . Assume that  $\sigma_e^2 = 0.25$ . The purpose of the measurement errors in this exercise is to break the stochastic singularity which could arise if we have only the technology shock in the model. You can treat  $\sigma_e^2$  as the known parameter.

### 1. Kalman filter.

- Using your log-linearization from the previous homework, write a routine that produces a state-space representation of the model for a given  $\theta$ . This state-space representation should consist of the reduced form solution of the rational expectations model and measurement equations (for  $x_t^*$ ). Note that in general for every choice of  $\theta$  you need to solve for steady state values of the variables. This step will affect your linearization. In this problem, you **do not** need to recompute the steady state (why?).
- Using assumed parameter values and your reduced-form solution to the model, simulate series for  $x_t^*$ . Set the sample size of the simulated series to  $T=1,000$  (don't forget about burn-in period).
- Using the state-space from step (a), estimate  $\theta$  by applying the Kalman filter to the  $x_t^*$  series. You can set initial values of  $\theta$  in the optimization routine equal to the true value of  $\theta$ . Compute the standard error for your estimate of  $\theta$  using the negative inverse of Hessian of the log-likelihood. For the purpose, you need to use `fminunc`,

not `fmincon`. The hessian matrix computed by `fmincon` may be incorrect because of the BFGS algorithm that `fmincon` adopts for speed.